

Pundarikaksh Narayan Tripathi

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Experience

FlyRank AI

Machine Learning Engineering Intern

Remote

July 2026 – Present

- Built a scalable ranking pipeline processing 79M+ interaction records using DuckDB and scikit-learn, engineering features for content discoverability and temporal engagement as part of a structured, project-based curriculum.
- Applied causal inference techniques to isolate confounding variables in user behavior data, using data-leakage-safe train/test splitting to support counterfactual analysis.
- Replicated the program's target benchmark for recommendation Precision@50 (0.24 → 0.74, a 208% relative improvement over heuristic baselines) by building a feature importance evaluation framework.

Projects

🔗 Causal-DML: Production Causal Inference Engine for OTT | PyTorch, DoWhy, EconML, Streamlit, DuckDB

- Engineered an enterprise-grade causal inference pipeline processing multi-gigabyte OTT streaming logs (KKBox) via DuckDB to estimate the Conditional Average Treatment Effect (CATE) of individualized retention interventions.
- Developed a Structural Causal Model (SCM) employing Double Machine Learning (EconML, DoWhy) backed by Random Forest nuisance estimators to orthogonally isolate the true causal impact of treatments on user churn.
- Deployed a real-time FastAPI inference backend paired with a dynamic Streamlit dashboard, enabling stakeholders to dynamically simulate counterfactual outcomes and evaluate long-term value (LTV) impacts across user cohorts. **Source:** github.com/PundarikakshNTripathi/Causal-DML

🔗 Aegis: Autonomous Agent Control Plane | Go, eBPF, Reinforcement Learning, Contextual Bandits, SQLite

- Designed a temporal behavior-graph anomaly scorer to track autonomous agent actions, mapping sequential behaviors into an SQLite-backed Approximate Nearest Neighbor (ANN) episodic memory layer.
- Utilized an offline-trained LinUCB contextual bandit algorithm, bounded by a Process Reward Model (PRM), to evaluate behavioral context and adapt threshold sensitivities dynamically without static signatures.
- Architected a zero-trust enforcement boundary using eBPF Linux Security Module (bpf_lsm) hooks, compiling AWS Cedar ABAC policies directly into kernel memory to block anomalous syscalls synchronously. **Source:** github.com/Quiet-Intelligence/aegis

🔗 TernixEngine: Hardware-Optimized Edge Inference | C++20, AVX2 SIMD, PyBind11, CUDA, Transformers

- Implemented a branchless 1.58-bit ternary matrix multiplication kernel using AVX2 intrinsics (`_mm256_srlv_epi32`, `_mm256_sign_epi32`) to bypass FP16 math and completely eliminate CPU branch mispredictions.
- Benchmarked an 8.2x throughput speedup (15.00ms vs 122.40ms baseline) for a 512x512 matrix execution on an i7-14650HX, mapping binary weights to sign controls within L0 registers.
- Integrated asynchronous warp tiling via `cp.async` and `__shfl_down_sync` in CUDA, binding the bare-metal C++20 engine to Python via PyBind11 and a 32-byte aligned tensor allocator. **Source:** github.com/Quiet-Intelligence/TernixEngine

🔗 nanoDist: Distributed ML Engine | NumPy, Distributed Systems, ZeRO Stage 2, Docker, MLOps

- Engineered a distributed training orchestration engine from scratch in pure NumPy, implementing peer-to-peer Ring All-Reduce primitives without relying on external communication frameworks.
- Slashed peak parameter memory by 68.07% (from 105.97 MB to 33.84 MB per GPU) through mathematical ZeRO Stage 2 optimizer state partitioning and FP16 static-scaled mixed precision.
- Bounded memory consumption during backpropagation via dynamic activation checkpointing, while containerizing the distributed cluster topology across isolated Docker instances. **Source:** github.com/PundarikakshNTripathi/nanoDist

Technical Skills

Languages: Python, SQL, C, C++ (C++17/20), CUDA C++, Triton, Go, Node.js

Modeling & Analysis: Causal Inference, Counterfactual Reasoning, Double Machine Learning, Sequential Modeling

Security & Kernel Systems: eBPF (Kernel Telemetry), Linux Security Modules, Zero-Trust Architectures, AWS Cedar (Formal ABAC), Vulnerability Research, Linux

ML & Reinforcement Learning: PyTorch, JAX, ONNX, EconML, DoWhy, scikit-learn, XGBoost, LightGBM, Transformers, Hugging Face, Reinforcement Learning (LinUCB Contextual Bandits), Process Reward Models (PRM), Quantization (1.58-bit / GGUF), OpenCV

Low-Level & Hardware: GPU Programming, Hardware Intrinsics (SIMD/AVX2), Distributed Data Parallel (DDP), Memory Management

Data, MLOps & Infrastructure: DuckDB, MLflow, Streamlit, Apache Arrow, Kafka, Apache Spark, Kubernetes (K8s), Optuna, CMake, Docker, FastAPI, gRPC, Redis, PostgreSQL, Elasticsearch, SQLite Vector Search, Microservices

Education

Babu Banarasi Das University

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Lucknow, UP

Expected August 2027

- **Relevant Coursework:** Data Science, Machine Learning, Linear Algebra, Calculus, Statistics, Probability, Data Structures, Design and Analysis of Algorithms, Database Management Systems, Computer Networking.

Achievements

- Secured **2nd Place at Masai x NoBroker HackArena 2025** for **CivicAgent**. Engineered an automated civic reporting platform featuring deterministic SLA tracking, transparent public dashboards, and scheduled escalation workflows to streamline municipal issue resolution.

Social Engagements & Leadership

Campus Partner: Perplexity AI (Fall '25 Cohort, Aug 2025 – Jan 2026)

Member: Google Developer Groups (GDG) Lucknow (Aug 2025 – Present)